

No-Go Evidence for Train-Free Black-Box Visual Hallucination Mitigation

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Abstract

Black-box vision-language systems can expose answer instability under alternate prompts, but a diagnostic signal is useful for hallucination mitigation only when it improves the final scored answer under the benchmark metric. We report a train-free GPT-5-series hosted-VLM evaluation of prompt-only mitigation candidates across HallusionBench text-only rows, MMIR image/document rows, and POPEv2 exact-image rows. The latest evidence-first entailment policy is a pilot no-go: it scores 0.3125 on MMIR versus 0.3750 for paraphrase consistency and 0.7500 on POPEv2 versus 0.8750 for paraphrase consistency, so it fails the family-wise stop rule. A full-scale selective-utility route is also a no-go under abstention credit 0.5 because POPEv2 utility is 0.8840 versus 0.8920 for forced paraphrase consistency. The older counterfactual gate is retained only as historical pilot context. The contribution is a calibrated negative result: prompt-only disagreement and evidence-audit signals are useful diagnostics in these artifacts, but they do not yet form a benchmark-improving mitigation policy or support visual HallusionBench claims.

Introduction

Vision-language models can answer visual questions fluently while grounding their responses poorly in the input image. POPE-style probing tests object hallucination (Li et al. 2023). HallusionBench targets language hallucination and visual illusion (Guan et al. 2023). MMIR evaluates multimodal inconsistency reasoning over image-rich sources (Yan et al. 2025). Recent mitigation work studies attention calibration, holistic reasoning, and training-free correction strategies (Fazli et al. 2025; Wu et al. 2024b; Anand et al. 2026). These threads motivate a narrower black-box question: when a hosted model exposes only text/image API calls, can prompt-only checks improve benchmark-scored answers rather than merely reveal instability?

The current answer from this project is negative. The latest method version, an evidence-first entailment policy, compares neutral prompting, paraphrase consistency, and an evidence-audit final-answer policy on a small paired pilot. It fails the stop rule on two visual benchmark families: MMIR scores 0.3125 for the policy versus 0.3750 for paraphrase consistency, and POPEv2 scores 0.7500 for the policy versus 0.8750 for paraphrase consistency. The pilot decision therefore blocks scale-up and blocks use of the policy as positive paper evidence.

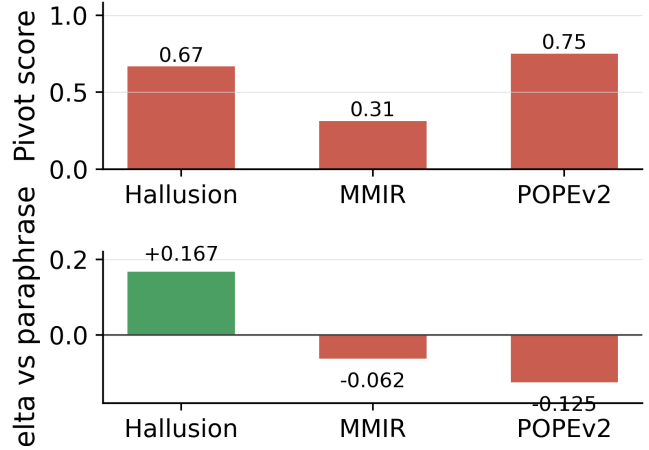


Figure 1: The current pilot stop-rule plot shows why the evidence-first pivot remains a no-go: MMIR and POPEv2 fail against the paraphrase comparator.

The powered selective-abstention route is also negative. On full-scale source rows, the selective utility condition improves HallusionBench text-only and MMIR relative to forced paraphrase consistency, but the declared stop rule is family-wise and POPEv2 fails: utility is 0.8840 while the forced paraphrase comparator is 0.8920 at abstention credit 0.5. That single-family failure blocks a full positive claim even though some aggregate summaries are tempting.

Figure 1 summarizes the current pilot stop-rule evidence. This paper therefore treats the older counterfactual gate as historical context: it also failed to justify a positive mitigation claim, but it is not the central method or contribution here.

This framing makes the paper intentionally narrower than a leaderboard report. The evaluated backend is fixed to a GPT-5-series hosted VLM in the executed rows. The scientific question is whether cheap black-box wrappers around that backend can pass a predeclared no-harm and family-wise improvement screen. The answer is negative under two action spaces: evidence-first prompting changes the final answer after an audit, while selective utility abstains on risk-flagged rows. Both actions reveal useful diagnostics, but neither produces the benchmark-wide improvement needed for a mitigation claim.

The main contribution is therefore an evidence boundary: the paper records which prompt-only signals were measured, where they failed, and why the failure should stop scale-up. This is useful because prompt-disagreement artifacts can otherwise be easy to overlook. A risk flag, an explanation-like audit, or an abstention utility gain on two families does not by itself establish hallucination mitigation. The benchmark decision must be tied to final scored answers under the stated family contract.

Related Work

Multimodal hallucination benchmarks. POPE evaluates object hallucination with binary object-presence questions (Li et al. 2023). HallusionBench broadens the target to language hallucination and visual illusion (Guan et al. 2023). MMIR targets image-rich inconsistency reasoning across documents, web pages, posters, slides, and tables (Yan et al. 2025). Additional benchmark work studies hallucination attribution, instruction-following failures, and multimodal robustness (Pham and Schott 2024; Wu et al. 2024a; Wang et al. 2026). The reported evidence uses these benchmark families as diagnostic measurement targets rather than as support for a broad deployment claim.

The benchmark mixture matters because the same wrapper can fail for different reasons. POPEv2 asks for a binary visual-presence judgment, so a conservative policy can lose by rejecting an actually visible object. MMIR gives partial credit for option IDs, so a verbose answer that emits extra IDs can lose even when it mentions one correct option. HallusionBench is treated here only through the text-only execution-ready rows available in the completed runs. Combining these families is still valuable, but the paper does not collapse them into one aggregate win criterion.

Inference-time and training-free mitigation. Training-free or inference-time hallucination methods can alter reasoning structure, calibrate visual evidence, or add verification stages (Kang et al. 2025a,b; Fazli et al. 2025; Zhang et al. 2026). CRoPS is especially relevant as a training-free hallucination mitigation framework (Anand et al. 2026). In this project, CRoPS remains a scope boundary because the completed runs use a hosted black-box interface rather than the compatible white-box generation interface required for a fair execution. This is not a CRoPS failure claim.

The current wrappers occupy a weaker but common operating point: no weights, no logits, no attention maps, and no task-specific training data. That constraint makes prompt-only checks attractive, but it also removes the mechanisms that many prior methods use to localize or correct visual grounding failures. The paper’s negative result should be read inside that constraint. It says the evaluated GPT-5-series hosted-VLM wrappers did not clear their contracts, not that richer white-box or trained methods cannot mitigate hallucination.

Prompt-induced failures and consistency checks. Prompt wording can steer visual-language systems away from image evidence (Khayatan et al. 2026; Yakun et al. 2026). Prompt-pair checks are attractive because they are cheap and API-compatible, but the current evidence reinforces a simple distinction: answer instability is a diagnostic

observation, not a mitigation result. A mitigation claim requires better final benchmark-scored answers.

Method

The evaluated system is a train-free black-box wrapper around a GPT-5-series hosted vision-language backend. The executed scored rows use the same hosted backend throughout, and the paper reports that family-level backend identifier rather than a private endpoint handle. Each task row is run through a fixed benchmark wrapper that records the question, parsed answer, score, and raw response. API inference uses no explicit temperature or top-p override; the runner sets only the maximum completion-token budget. The wrapper has no model-internal access, no weight updates, no retrieval, no learned reranker, and no post-hoc metric override.

Three current pilot conditions define the evidence-first policy. *Neutral prompting* asks the benchmark question directly. *Paraphrase consistency* asks a semantically equivalent variant and serves as the feasible prompt-only comparator. *Evidence-first entailment* asks the model to audit visible or textual evidence before emitting the final benchmark answer.

The policy is intentionally simple enough to audit. For binary POPEv2 and HallusionBench rows, the output is parsed to a yes/no answer. For MMIR, the output is parsed to one or two answer IDs, and outputs with unsupported extra IDs are scored by the benchmark wrapper rather than manually corrected. This matters because some failures are parsing-valid but answer-invalid: the evidence audit can sound plausible while still selecting the wrong option or emitting too many IDs.

The powered selective-abstention route is a separate evaluated policy. It combines full-scale neutral and paraphrase source rows with a selective utility score. Abstained rows receive credit 0.5 by contract, answered rows receive the underlying answer score, and the route must clear every benchmark family against forced paraphrase consistency. It does not clear that rule because POPEv2 is below the comparator.

The older counterfactual gate is not discarded, but it is demoted. That earlier pilot compared neutral prompting, paraphrase consistency, and a premise-challenging gate. Its result was also negative, so it now functions only as historical evidence that simple prompt-pair disagreement was not enough to justify scale-up. It is not used as a positive baseline or as the central manuscript finding.

Experimental Setup

The evaluation covers four condition sets. The older counterfactual pilot covers 12 HallusionBench text-only rows, 10 MMIR rows, and 20 POPEv2 rows per condition. The evidence-first pilot covers 6 HallusionBench text-only rows, 8 MMIR rows, and 8 POPEv2 rows per condition with a 160-token maximum completion budget. The powered source comparison covers 178 HallusionBench text-only rows, 499 MMIR rows, and 500 POPEv2 rows for neutral and paraphrase prompting with a 96-token maximum completion budget. The powered selective-utility condition is postprocessing-only and covers the same full-scale family sizes.

HallusionBench remains text-only scoped in all completed paper-facing evidence. The recorded pivot scope excludes visual HallusionBench evidence, so the manuscript makes no visual HallusionBench improvement or failure claim. MMIR and POPEv2 are visual/image-row evidence, but their current role is diagnostic: they show why the latest pivot and powered utility route should not be promoted as successful mitigation.

Metrics follow the benchmark wrappers. HallusionBench text-only and POPEv2 use accuracy. MMIR uses partial-credit score. Selective utility reports the abstention-credit objective with coverage and abstention rate. All inference rows use the same GPT-5-series hosted VLM backend; this paper reports paired policy comparisons rather than a public-checkpoint leaderboard claim.

The row selection is deliberately transparent. The evidence-first pivot is a small paired pilot, so it is allowed only to reject or motivate a method, not to support a final success claim. The powered source rows are larger and can test the selective-utility decision, but they inherit the declared utility contract: abstention credit is fixed at 0.5 for the headline decision, and the family-wise rule cannot be cleared by averaging over the POPEv2 failure. The sensitivity table later reports other abstention credits as analysis only.

Results

Table 2 is the central manuscript result. It places the pilot evidence-first policy beside the full-scale selective-utility route so the no-go conclusions can be checked against benchmark source, task count, model family, policy role, decoding budget, metric, and comparator in one matrix. The pilot fails because MMIR and POPEv2 are below the feasible paraphrase comparator; the powered route fails because POPEv2 utility is below forced paraphrase.

The HallusionBench text-only row does not rescue the pivot. It ties neutral at 0.6667 and beats paraphrase on that narrow text-only slice, but the stop rule is family-wise and visual HallusionBench evidence is absent. The current evidence therefore cannot be written as a visual HallusionBench claim. MMIR and POPEv2 are the visual/image-row checks in this pivot, and both are below paraphrase consistency: MMIR is 0.3125 versus 0.3750, and POPEv2 is 0.7500 versus 0.8750. The POPEv2 decision is additionally constrained by no-harm preservation: the pivot decision records seven neutral no-harm candidates and only six preserved under the evidence-first condition, while the positive no-harm subset is preserved. The failure is therefore not merely low average score; it is a violation of the predeclared family and no-harm checks.

Figure 2 visualizes the same no-go result. The important reading is not a pooled mean; it is the absence of a clean family-wise scale-up case under the declared comparison.

Figure 3 gives the full-scale selective utility result. HallusionBench text-only utility is 0.7556 versus forced paraphrase 0.7416, and MMIR utility is 0.5471 versus 0.5050, but POPEv2 is the blocking family with 0.8840 versus 0.8920. Because the declared rule requires every family to clear the comparator, this route remains a full-scale no-go rather than a positive mitigation result.

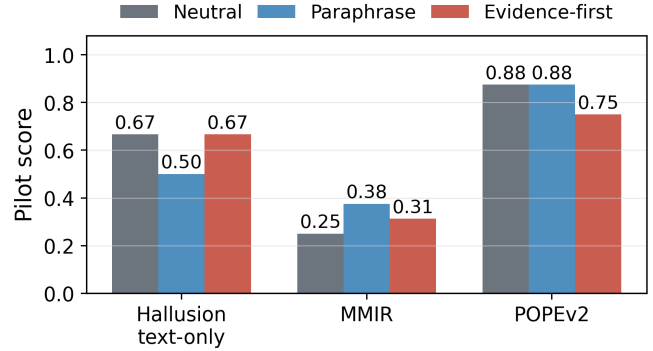


Figure 2: Pilot scores by benchmark show the current evidence-first pivot has no family-wise scale-up case once MMIR and POPEv2 are compared with paraphrase consistency.

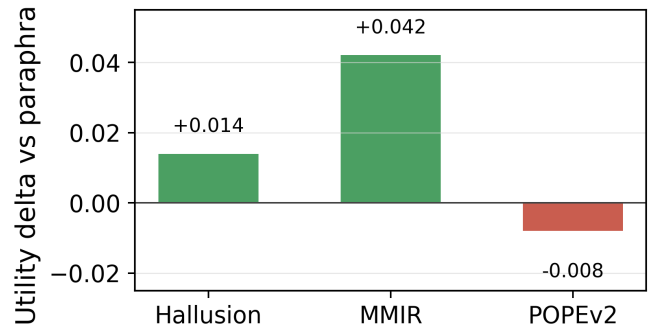


Figure 3: The powered selective-utility run is also a no-go: POPEv2 utility is 0.8840, below the forced paraphrase comparator at 0.8920 under abstention credit 0.5.

The powered result also shows why coverage alone is not enough. POPEv2 has the highest coverage, 0.9280, and answered-row accuracy, 0.9138, but its utility still trails forced paraphrase by 0.0080. MMIR has lower coverage, 0.5711, yet gains utility because abstention removes enough risk-flagged rows under the 0.5 credit contract. Appendix Tables 4 and 5 separate credit sensitivity from the fixed-credit breakdown. The body keeps the decision point: coverage can explain selective utility, but it cannot override a family-wise POPEv2 failure.

This matters for the claim boundary. A cross-family utility mean would make the route look favorable. The declared rule instead asks whether every family clears forced paraphrase at the same credit. MMIR and HallusionBench text-only gains are real under the utility definition, but they are insufficient because POPEv2 is the family most directly tied to binary object hallucination and remains below the comparator.

The utility sensitivity sweep is supportive rather than central, so it is reported in Appendix Table 4. A higher abstention credit can make every family clear, but that is not the predeclared headline contract. At 0.75 credit the policy rewards abstention strongly enough that POPEv2 becomes positive; at 0.5 it remains negative. The paper therefore reports the 0.5 no-go and treats the sensitivity only as evidence about the

Benchmark/source	Rows	Model/backend	Method or baseline role	Budget	Metric	Decision takeaway
HB text-only source rows	6	GPT-5-series hosted VLM	Evidence-first pilot versus neutral/paraphrase	160 tok.; no temp/top-p override	accuracy	Text-only pilot only; not visual evidence.
MMIR image/document rows	8	GPT-5-series hosted VLM	Evidence-first pilot versus neutral/paraphrase	160 tok.; no temp/top-p override	partial	Trails paraphrase, so pilot is no-go.
POPEv2 exact-image rows	8	GPT-5-series hosted VLM	Evidence-first pilot versus neutral/paraphrase	160 tok.; no temp/top-p override	accuracy	Trails paraphrase, so pilot is no-go.
HB text-only source rows	178	GPT-5-series hosted VLM	Selective utility versus forced paraphrase	96 tok. source; utility postprocess-ing only	utility	Above paraphrase, but not sufficient.
MMIR image/document rows	499	GPT-5-series hosted VLM	Selective utility versus forced paraphrase	96 tok. source; utility postprocess-ing only	utility	Above paraphrase, but not sufficient.
POPEv2 exact-image rows	500	GPT-5-series hosted VLM	Selective utility versus forced paraphrase	96 tok. source; utility postprocess-ing only	utility	Blocking family at credit 0.5.

Table 1: The setup matrix exposes the evaluated backend and budget for each paper-facing route; both active routes remain no-go because each fails at least one visual/image-family comparator.

utility definition.

Diagnostic Analysis

The pivot failure taxonomy is useful as future-method motivation. It is not a result showing that the current pivot works. The taxonomy contains 11 failed or partial pivot rows: four missing-visual-evidence cases, four multi-ID over-emission cases, two other observed response patterns, and one over-conservatism case. Appendix Table 6 gives the compact count table; the body uses the taxonomy only to explain where the evidence-first prompt fails to become a reliable final-answer policy.

MMIR failures mostly involve answer selection and partial-credit structure. Several responses parse cleanly but select wrong option IDs, and several emit extra IDs beyond the gold set. For example, the taxonomy records cases where the answer contains one plausible option plus an unsupported extra option, which converts a near miss into only partial credit or zero credit under the MMIR wrapper. This pattern suggests that a future method may need structured option verification rather than another global prompt wrapper.

POPEv2 exposes a different failure mode. One error is over-conservative: the object is present, but the response rejects it as not directly visible. Another error asserts visible evidence for an absent object. This is a sharper failure than an explanation-quality issue because POPEv2 scoring depends on the final yes/no answer. The lesson is not that skepticism works or fails universally; it is that a useful binary policy must distinguish when visual evidence should override the skeptical default.

HallusionBench should remain separate in interpretation. The completed HallusionBench evidence is text-only execution-ready rows. It can support a text-only diagnostic statement, but not a visual HallusionBench result. Any future visual HallusionBench claim requires official visual assets

and a fresh visual run.

The taxonomy also explains why the powered selective route was tempting. Risk-flagged rows correlate with many errors in HallusionBench text-only and MMIR, so abstention can improve those utility scores at credit 0.5. The same mechanism does not clear POPEv2, where the risk rate is lower and the forced paraphrase comparator is already strong. A future policy would therefore need a family-aware action after risk detection, not just a global abstain-or-answer rule.

Discussion and Limitations

The current manuscript is a negative diagnostic paper, not a positive mitigation paper. The supported claims are narrow: evidence-first entailment is a pilot no-go; selective utility is a full-scale no-go under abstention credit 0.5; the older counterfactual gate is historical negative context; the MMIR/POPEv2 taxonomy is future-method motivation; and HallusionBench remains text-only scope. The body omits the unsupported readings: no successful mitigation claim, no visual HallusionBench result, no scale-up claim, and no CRoPS failure claim under this black-box interface.

The older counterfactual gate helps explain the research trajectory but not the current central claim. That route already indicated that prompt-pair disagreement was not enough to justify scale-up. The later evidence-first pivot tested a different action and still did not clear MMIR or POPEv2. The powered route then tested selective abstention at larger scale and still failed POPEv2 under the declared credit. Across the three routes, the artifacts identify method bottlenecks rather than a deployable mitigation.

The negative result narrows the next experiment. The failure is not caused by missing scored rows for the reported policies: the rows retain responses, parsed answers, gold fields where adapters provide them, and family summaries. The missing evidence is a successful action policy. A succes-

Benchmark/source	Rows	Model/backend	Policy role	Budget	Metric	Neutral	Paraphrase	Policy score and decision
HB text	6	GPT-5-series hosted VLM	Evidence audit pilot	160 tok.; no temp/top-p override	accuracy	0.6667	0.5000	0.6667 ; text-only only.
MMIR	8	GPT-5-series hosted VLM	Evidence audit pilot	160 tok.; no temp/top-p override	partial	0.2500	0.3750	0.3125; trails paraphrase.
POPEv2	8	GPT-5-series hosted VLM	Evidence audit pilot	160 tok.; no temp/top-p override	accuracy	0.8750	0.8750	0.7500; trails paraphrase.
HB text	178	GPT-5-series hosted VLM	Selective utility full	96 tok. source; postproc.	utility	0.7697	0.7416	0.7556; above paraphrase.
MMIR	499	GPT-5-series hosted VLM	Selective utility full	96 tok. source; postproc.	utility	0.4579	0.5050	0.5471 ; above paraphrase.
POPEv2	500	GPT-5-series hosted VLM	Selective utility full	96 tok. source; postproc.	utility	0.8760	0.8920	0.8840; blocking family.

Table 2: The review matrix exposes both no-go routes: the evidence-first pilot loses to paraphrase on MMIR and POPEv2, and the powered selective-utility route loses to forced paraphrase on POPEv2 under abstention credit 0.5.

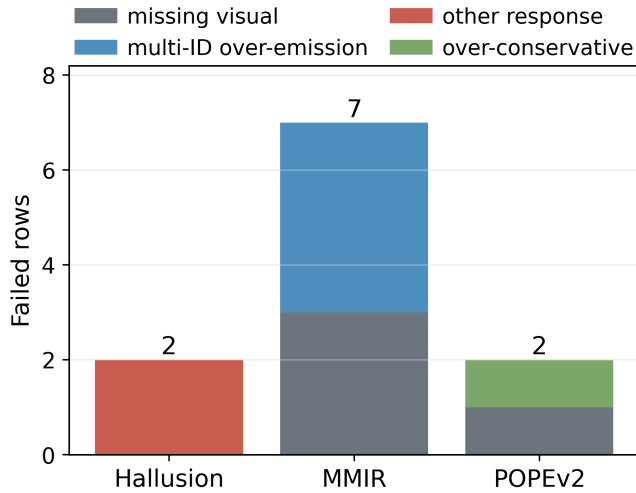


Figure 4: The MMIR/POPEv2 error taxonomy is diagnostic only: missing visual evidence, multi-ID over-emission, and over-conservatism explain failures but do not support a positive method claim.

or train-free method should predeclare a task-aware action after risk detection, preserve no-harm rows, compare against neutral prompting and paraphrase consistency, and clear every benchmark family before restoring a positive paper claim.

The comparator choice is intentionally conservative for a black-box study. Neutral prompting measures the direct benchmark answer from the same hosted backend. Paraphrase consistency is a feasible same-backend, same-budget, prompt-only alternative. White-box training-free methods, model-editing methods, and approaches that need hidden states or logits are important related work, but they are not

fair baselines for the completed hosted-backend runs unless the experiment is rerun in a compatible setting. Conversely, the present no-go should not be read as evidence that those methods fail.

The limitations are substantive. The evidence-first pivot is small by design: it can reject a candidate action but cannot support a broad conclusion about all prompt-only mitigation. The powered selective-utility route is larger, but it depends on the utility contract. At credit 0.5, POPEv2 blocks the family-wise rule; at higher credit, abstention becomes more valuable, but that is a different contract. The evaluated backend is a GPT-5-series hosted black-box VLM, so the study is relevant for API-only users but cannot support attention, logit, hidden-state, or visual-module analysis. The observed failures are final-answer phenomena, not mechanistic causal attributions.

These constraints are the paper’s main safeguard against overinterpretation. A risk flag, an evidence-audit explanation, or a positive cross-family mean is not a mitigation result unless the final benchmark-scored answers satisfy the declared family contract. The body therefore reports negative evidence without tuning the metric after the fact. The abstention-credit sensitivity is shown as analysis, but the headline remains credit 0.5. HallusionBench text-only rows are reported as text-only rows, not reinterpreted as visual evidence. These choices make the paper less flattering, but they make the no-go conclusion reproducible and comparable.

Claim Scope and Evidence Integrity

The supported claims are deliberately narrow. The evidence-first policy is a pilot no-go, the selective-utility route is a full-scale no-go under abstention credit 0.5, the older counterfactual gate is historical negative context, the MMIR/POPEv2 taxonomy is diagnostic future-method motivation, and HallusionBench is text-only scope. Numeric claims are

reported through benchmark tables, figures, and stop-rule prose.

The predeclared stop rules constrain how broad the conclusion can be. The evidence-first policy does not justify scale-up because it fails MMIR and POPEv2 against paraphrase consistency; selective utility does not clear the family-wise rule at credit 0.5 because POPEv2 remains below the forced comparator. Those observations govern the claim scope: no scale-up, no successful method claim, and no positive result can be inferred from the present evidence. The paper also avoids the inverse overclaim. It does not argue that all train-free mitigation is impossible, that the hosted VLM cannot be improved by any wrapper, or that selective abstention is useless. It records that the measured wrappers did not satisfy their own contracts.

The no-go conclusion depends on scored benchmark outputs rather than on a post-hoc narrative judgment. The evaluated rows retain model responses, parsed answers, benchmark gold fields where available, and family summaries. This matters because a negative result is informative only if the failure is measured under the same wrappers that would have credited a success.

The pivot and powered routes fail for different reasons. The pilot directly changes final answers and loses visual/image-family comparisons. The powered route abstains selectively and still cannot clear POPEv2 utility. The POPEv2 pivot loses one neutral-safe row, and the powered route has high POPEv2 coverage but still trails forced paraphrase. These are method-level failures, not simply low aggregate scores.

The benchmark coverage is intentionally bounded. POPEv2 and MMIR provide visual/image-row evidence in the completed runs, while HallusionBench remains text-only. The paper therefore uses HallusionBench to discuss text-only diagnostic behavior and not visual hallucination mitigation. A future visual HallusionBench study would need official visual assets, fresh execution, and a preregistered evaluation claim before it could be mixed with the present evidence. That boundary is especially important because the HallusionBench text-only slice is the easiest row to overread: it can show that a policy did not damage a small text-only subset, but it cannot validate visual grounding.

Comparison Fairness

The comparator choice is intentionally conservative for a black-box study. Neutral prompting measures the direct benchmark answer from the same hosted backend. Paraphrase consistency changes wording but not model access, task data, or scoring. The evidence-first pivot and the powered selective-utility policy must beat or tie that feasible comparator on every benchmark family before they can be considered useful.

The paper does not compare against methods that require incompatible access. White-box training-free methods, model-editing methods, or approaches that need hidden states and logits are important related work, but they are not fair baselines for the completed hosted-backend runs unless the experiment is rerun in a compatible setting. Conversely, the negative result should not be used to claim those methods fail.

The comparison is restricted to policies that can be applied to the same row set, through the same hosted backend, under the same benchmark parsers.

The powered selective route is the clearest fairness stress test. It has full-scale neutral and paraphrase source rows and can abstain on risk-flagged cases. That could look attractive if the paper reported only cross-family mean utility. The family-wise rule prevents that shortcut: a policy that improves MMIR by abstaining on many risk rows must still avoid damaging POPEv2, where the comparator is already strong and the abstention rate is low.

The pilot route has a different fairness issue. It is too small to prove general success, but it is large enough to reject the specific action under the declared rule because failures occur on both visual/image families. Reporting it as a no-go is therefore not a pessimistic reading of a weak result; it is the planned use of a pilot stop rule. A larger run would be justified only after a method clears the pilot conditions or after the operator explicitly changes the scientific objective.

This negative framing also avoids a common failure mode in hallucination-mitigation papers: treating explanation quality as equivalent to answer quality. The evidence-first prompt often produces plausible evidence language, but the benchmark only credits the final answer. If the final answer overemits IDs, rejects a visible object, or asserts evidence for an absent object, the explanation does not rescue the method. The main lesson is that audit-like text must be evaluated by final benchmark outcomes, not by whether it sounds grounded.

Successor Checks

A successor experiment should treat the present evidence as a preflight screen before spending another full run. The first check is MMIR answer control. The current evidence-first action loses partly because the response can emit extra option IDs or select an option unsupported by the image/document context. A new train-free wrapper should verify the option set before the final answer is parsed and report whether the over-emission rows are repaired before claiming a broader score improvement.

The second check is POPEv2 no-harm preservation. The pilot decision records a loss on a neutral-safe row, and the powered selective-utility result shows that high coverage does not guarantee a utility win when the forced paraphrase comparator is already strong. A future method should separately report positive-object preservation, neutral-safe preservation, and family utility rather than only a cross-family mean. If the method cannot preserve these rows, the stop-rule result should remain negative even if another family improves.

The third check is benchmark-scope hygiene. HallusionBench can enter this manuscript only through the completed text-only rows, so a future visual HallusionBench claim requires official visual assets, fresh scored rows, declared splits, metrics, and evaluation scope before it is mixed with MMIR and POPEv2 evidence. This prevents a text-only diagnostic improvement from being rebranded as visual hallucination mitigation.

The fourth check is backend and comparator compatibility. The completed comparison is same-backend and

Check	Measured bottleneck	Required evidence before scale-up
MMIR option control	Evidence-first rows can over-emit IDs or choose unsupported IDs.	Option verification that repairs those rows under the official parser.
POPEv2 no-harm	The pivot loses a neutral-safe row, and utility trails forced paraphrase.	Positive-object and neutral-safe preservation plus family utility.
HB visual scope	Completed HallusionBench rows are text-only.	Official visual assets, fresh visual rows, and declared split/metric scope.
Comparator compatibility	Current evidence is same-backend black-box only.	Compatible baselines whenever backend access or action space changes.

Table 3: Successor checks convert the no-go evidence into concrete preflight requirements; failing any row keeps the paper in diagnostic rather than positive-mitigation scope.

same-wrapper: neutral prompting, paraphrase consistency, evidence-first prompting, and selective utility all operate through the hosted black-box setting. If a future study introduces an open-weight or white-box method, it should be treated as a new experiment with compatible baselines, not as a retroactive explanation of the present no-go. Conversely, if the study remains hosted and black-box, the fair comparator is still a same-backend prompt-only route that can run on the same rows and under the same parsers.

These checks are procedural because the present result is a stop-rule artifact. They do not turn the failed policies into a positive contribution. They specify what evidence would have to change for a future paper to make a positive claim without weakening the benchmark contract that produced the current no-go.

Reviewer Interpretation

The most important review distinction is between diagnostic signal and mitigation action. The evidence-first prompt produces an audit before answering, and the selective route produces a risk flag before deciding whether to abstain. Both signals can be informative. Neither is credited by the benchmark unless the final answer or utility decision improves under the declared comparator. This is why the paper keeps the result negative even when the intermediate behavior looks useful.

The pilot evidence should be read as a stop decision, not as a final estimate of every prompt-only method. Its role is to test whether the current action is worth scaling. It fails that role because both visual/image families are below paraphrase consistency. A larger run of the same action would therefore be hard to justify without first changing the action space that caused the measured failures.

The powered utility evidence has a different review risk. It is large enough to make the selective-abstention result tempting, and two families improve at abstention credit 0.5.

The all-family rule prevents that partial gain from becoming the headline. POPEv2 remains below forced paraphrase, so the route is not a successful hallucination-mitigation policy under the declared utility contract.

The taxonomy supports interpretation but not rescue. Missing visual evidence, multi-ID over-emission, and over-conservatism are useful because they identify where a future wrapper should intervene. They do not retroactively improve the scored rows. The paper therefore uses the taxonomy as a design constraint for follow-up work rather than as evidence that the present wrapper is close enough to success.

This interpretation also limits what a reviewer should demand from the current manuscript. The fair correction is not to relabel the no-go as a win, average away POPEv2, or treat text-only HallusionBench rows as visual evidence. The fair next experiment would modify the action policy, keep the same comparator discipline, and rerun the affected benchmark families under a predeclared rule.

Conclusion

The current evidence supports a measured no-go result. Evidence-first visual entailment fails its pilot stop rule because MMIR and POPEv2 trail paraphrase consistency. Powered selective-abstention utility fails its full-scale stop rule because POPEv2 utility is lower than forced paraphrase consistency under abstention credit 0.5. HallusionBench remains text-only scope, and MMIR/POPEv2 error taxonomy rows motivate a future method rather than validating the current one.

The central lesson is methodological. Prompt-only audits can produce useful disagreement signals, plausible evidence statements, and informative risk flags, but those signals must be judged by final benchmark-scored answers. In these runs, the evidence-first action did not preserve the visual/image-family comparisons, and selective abstention did not clear the declared all-family utility rule. That combination rules out a positive mitigation claim while still preserving a useful diagnostic record for future work.

The next scientific step is therefore not to reinterpret the existing rows. A successor policy should change the action space: for MMIR, it should verify option IDs rather than rely on a global audit prompt; for POPEv2, it should distinguish skeptical rejection from genuine absence so that no-harm rows are preserved; for HallusionBench, it should add an official visual-asset execution before making visual claims. Until such a policy clears neutral and paraphrase comparators under a predeclared family-wise contract, the honest conclusion is diagnostic no-go evidence.

This also changes how the result should be used by later experiments. The evaluated rows should serve as a regression test for proposed wrappers: a new method should be checked first on the cases where evidence-first over-emits MMIR IDs, where POPEv2 loses a neutral-safe object-presence case, and where selective abstention fails to beat forced paraphrase at credit 0.5. Passing those targeted cases would not by itself prove success, but failing them would quickly reveal that the successor method has not addressed the measured bottleneck. The paper therefore leaves a concrete bar for follow-up work rather than a vague call for larger evaluation.

The no-go framing is also the appropriate stopping point for this train-free setting. The completed runs used a fixed hosted backend, fixed row scopes, fixed parsers, and fixed utility contracts. Changing the backend, adding white-box access, rerunning HallusionBench with visual assets, or changing the abstention reward could all be valuable future studies, but each would be a new experiment. Under the evidence available here, the measured answer is stable: the tested prompt-only actions expose diagnostic signals but do not yet improve the benchmark outcomes required for a hallucination-mitigation claim.

For a reviewer, this is the practical contribution. The paper turns a sequence of tempting partial signals into a falsifiable boundary: a small text-only improvement cannot stand in for visual evidence; a positive mean cannot stand in for all-family utility; and an evidence-sounding audit cannot stand in for a correct parsed answer. These boundaries are easy to lose in prompt-engineering studies because the artifacts look semantically rich even when the benchmark answer is wrong. Reporting the boundary explicitly makes the negative result reusable: later work can keep the same comparator discipline, reuse the failure taxonomy as a preflight check, and decide before scale-up whether a new train-free action has actually changed the measured bottleneck.

The paper should therefore be read as a stop-rule artifact with scientific content, not as a failed attempt to force a positive result. Its value is the measured refusal to promote a wrapper when the family-wise evidence says no.

That refusal is especially important because every tempting positive reading has a corresponding measured counterexample. The evidence-first route has an appealing audit structure, but MMIR and POPEv2 still trail paraphrase consistency. The selective route has an appealing utility story, but POPEv2 remains below forced paraphrase at the declared credit. The taxonomy has actionable failure categories, but those categories explain why a future action may be needed rather than showing that the current action works. Keeping those distinctions visible is the main contribution of this draft.

The same boundary should guide any future rescope. If the objective becomes a public-backbone comparison, the experiment should be rebuilt around compatible white-box or open-weight baselines. If the objective remains hosted black-box mitigation, the next method should first repair the observed MMIR over-emission and POPEv2 no-harm failures before scale-up. If the objective becomes abstention, the abstention credit should be declared before scoring. These are different studies, and the present manuscript keeps them separate.

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Reproducibility Checklist

Scope. This manuscript reports negative and diagnostic evidence. It does not report a successful mitigation method or a visual HallusionBench evaluation.

Data and benchmarks. The benchmark families are HallusionBench text-only execution-ready rows, MMIR official ready image/document rows, and POPEv2 repaired exact-image rows. HallusionBench visual claims are excluded.

Methods and baselines. The current pilot compares neutral prompting, paraphrase consistency, and evidence-first entailment. The powered utility route compares selective utility with forced paraphrase consistency under abstention credit 0.5. The counterfactual gate is older pilot context only.

Metrics. HallusionBench text-only and POPEv2 use accuracy. MMIR uses partial-credit score. Selective utility uses the predeclared abstention-credit objective.

Compute and training. The completed inference runs use the same GPT-5-series hosted vision-language backend. The pivot uses a 160-token maximum completion budget, powered source rows use a 96-token maximum completion budget, and the selective-utility run is postprocessing-only. No finetuning, weight update, learned reranker, simulator override, or metric override is part of the reported evidence.

Stopping rule. The evidence-first pivot is not scaled because MMIR and POPEv2 fail against paraphrase consistency. The powered utility route is not claimed positive because POPEv2 fails the family-wise utility rule.

Supplementary Evidence Boundaries

The current pivot fails the pilot stop rule, so scale-up and a positive mitigation claim are not supported.

The powered selective-utility condition is a full-scale no-go because POPEv2 utility is 0.8840 versus forced paraphrase 0.8920.

The older counterfactual pilot is kept to document an earlier failed prompt-pair route, not to center the manuscript claim.

The pivot taxonomy should be read as diagnostic evidence. It identifies observed failure modes that could guide a future policy, but those modes do not demonstrate improved benchmark performance.

Credit	HB text	MMIR	POPEv2	All pass?
0.00 delta	-0.0730	-0.1723	-0.0440	no
0.40 delta	-0.0034	-0.0008	-0.0152	no
0.50 delta	+0.0140	+0.0421	-0.0080	no
0.60 delta	+0.0315	+0.0850	-0.0008	no
0.75 delta	+0.0576	+0.1493	+0.0100	yes

Table 4: Credit-sweep deltas show the no-go boundary: POPEv2 remains negative at the declared credit 0.5, while the higher-credit positive row is analysis only.

Credit 0.5 row	HB text	MMIR	POPEv2
Coverage	0.8258	0.5711	0.9280
Abstention	0.1742	0.4289	0.0720
Answered acc.	0.8095	0.5825	0.9138
Utility	0.7556	0.5471	0.8840

Table 5: Fixed-credit 0.5 utility breakdown shows why POPEv2 blocks the headline: high coverage and answered accuracy still produce utility below forced paraphrase.

Supplementary Review Boundaries

This appendix preserves audit detail that is useful for review but not necessary for the main-body argument. The first boundary is claim scope. The pivot evidence rejects the evidence-first action because both visual/image families trail paraphrase consistency. It does not show a general failure of prompt-only mitigation. The selective-utility evidence rejects the declared credit-0.5 policy because POPEv2 remains

below forced paraphrase. It does not reject all abstention policies, because the sensitivity rows change when the utility contract changes.

The second boundary is comparison fairness. The body compares policies that can run through the same black-box hosted backend, benchmark adapters, and row sets. This excludes white-box baselines from the completed comparison because those methods require different model access, not because they are scientifically unimportant. A fair future comparison would need an open-weight execution of those methods or a hosted-backend wrapper with equivalent action space and budget.

The third boundary is benchmark scope. HallusionBench appears in the completed evidence only through text-only execution-ready rows. Those rows are useful for checking whether prompt disagreement or abstention flags correlate with answer quality in text-only cases, but they cannot support a visual HallusionBench hallucination claim. The present visual evidence comes from MMIR image/document rows and POPEv2 exact-image rows. Together, these boundaries block the attractive unsupported readings: a text-only HallusionBench slice is not visual evidence, a positive utility mean is not an all-family pass, and a failure taxonomy is not a positive mitigation result.

Supplementary Failure-Mode Use

The failure taxonomy is included to make the negative result actionable. Missing visual evidence cases mean the audit can select or describe evidence that is not sufficient for the benchmark answer. Multi-ID over-emission cases mean the model may name one plausible option while adding an unsupported option that the MMIR wrapper must score as partial or wrong. Over-conservatism means a skeptical visual policy can reject an object that is actually present. These are different error mechanisms, so a successor method should not be evaluated only by aggregate score after development; it should first be checked against the failure categories that caused the current route to stop.

The selective-utility route gives a second diagnostic use case. When abstention removes enough risk-flagged rows, utility can improve on some families, but the POPEv2 result shows that this is not automatically a mitigation. A binary object-presence task with a strong forced comparator can lose utility even when coverage is high. That is why the paper reports coverage, abstention rate, answered-row accuracy, utility, and delta together rather than treating abstention as a uniformly positive intervention.

Supplementary Decision Trace

The paper’s decision trace has three stops. The older counterfactual gate remains historical because it did not establish a positive scale-up case. The evidence-first pivot is the current direct-answer pilot and stops because MMIR and POPEv2 trail paraphrase consistency. The powered selective route is larger and postprocessing-based, but it stops because POPEv2 utility is below forced paraphrase under the declared credit. This ordering matters: the manuscript is not selecting the least favorable result after the fact. It reports the latest pivot

Failure type	Rows	Interpretation
Missing visual evidence	4	Audit selects unsupported or absent visual evidence.
Multi-ID over-emission	4	MMIR answer includes extra IDs beyond gold.
Other observed pattern	2	Text/context answer contradicts gold.
Over-conservatism	1	Visible object rejected as uncertain.

Table 6: The pivot taxonomy identifies concrete failure modes, but the counts are diagnostic rather than evidence of mitigation success.

and the powered route because those evaluated conditions and stop rules define the evidence boundary.

The decision trace also explains why the appendix keeps some audit tables. They let a reader reconstruct what kind of future evidence would change the conclusion. A new result would need a different action policy, a compatible backend and baseline plan, official visual HallusionBench execution for any visual HallusionBench claim, and a predeclared utility contract for any abstention headline. Without those changes, the present rows should remain a no-go diagnostic record.

Supplementary Audit Use

The appendix evidence is intended to support replication of the paper’s negative decision, not to introduce a second result. A reader can use the sensitivity table to check whether the utility conclusion depends on the abstention reward, the selective-utility table to see why POPEv2 blocks the headline contract at credit 0.5, and the taxonomy table to identify which final-answer failures a successor policy must repair. These checks are deliberately separated because they answer different review questions.

The first review question is whether the negative result follows from the declared contract. It does. The direct pilot contract requires the evidence-first final-answer policy to clear the visual/image families against paraphrase consistency, and it does not. The powered utility contract requires every family to clear forced paraphrase at the declared abstention credit, and POPEv2 does not. The appendix tables expose the supporting details without changing those body decisions.

The second review question is whether the result is merely an artifact of a conservative utility setting. The sensitivity rows show why that interpretation is incomplete. A higher abstention credit can make selective utility look favorable, but that is a different policy value judgment. The headline uses credit 0.5 because that was the contract for the powered decision. A future abstention paper could choose a different credit, but it would need to declare that credit before scoring.

The third review question is whether the failure taxonomy should count as positive evidence. It should not. The taxonomy is useful because it names observable bottlenecks: missing visual evidence, multi-ID over-emission, and over-

conservatism. Those labels make the next experiment easier to design, but they do not alter any parsed answer or benchmark score in the completed rows. The current policy therefore remains stopped even though the failure modes are interpretable.

The fourth review question is whether the paper hides a stronger baseline obligation. It does not claim a public-checkpoint leaderboard result. The completed comparisons are same-backend black-box wrappers with the same row sets and parsers. Methods that need logits, hidden states, visual-module access, or model weights are methodologically relevant but not directly comparable to these completed hosted-backend rows. A future study can broaden the baseline set only by changing the execution setting and rerunning compatible comparisons.

These audit uses are intentionally modest. They make the no-go reproducible as a decision, clarify why tempting partial signals are not enough, and preserve the exact follow-up requirements. They do not convert a failed wrapper into a method contribution, and they do not weaken the requirement that future positive claims clear the family-wise comparator under a predeclared contract.